

XGBoost Challenger Model Feature Importance & SHAP Report

Model Name: XGBoost Default Predictor v1.0
Evaluation Target: Feature Importance and Risk Drivers

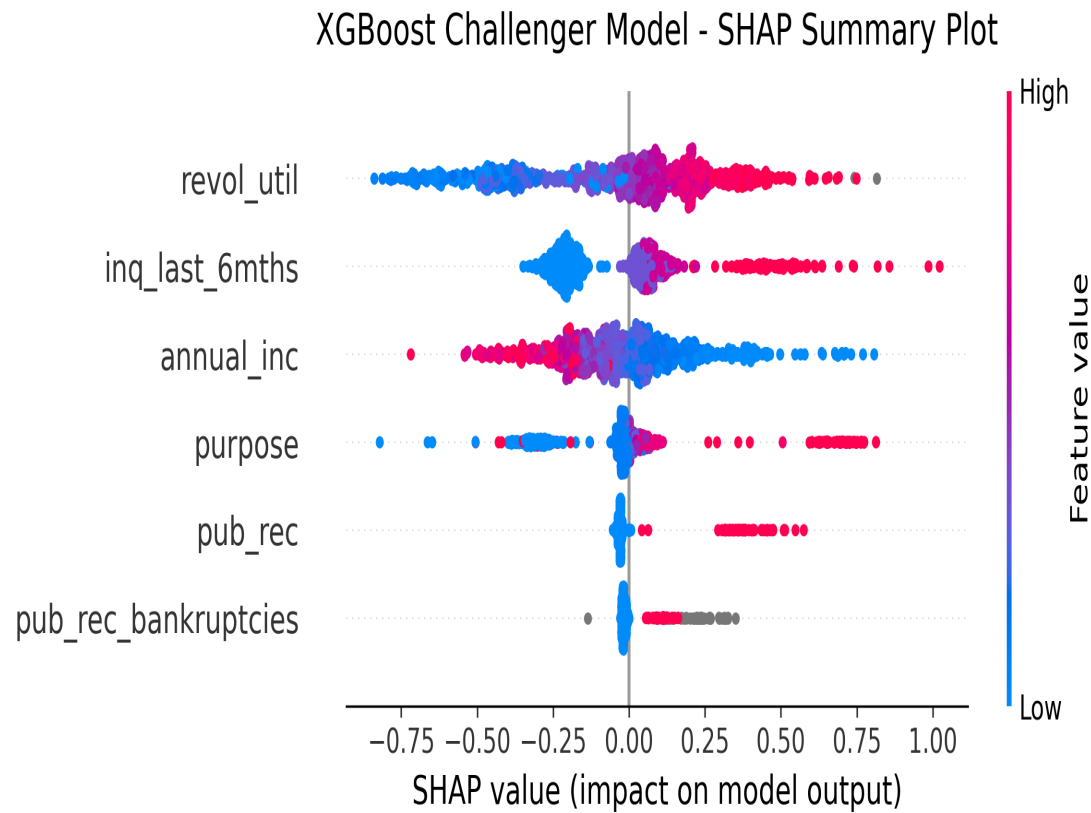
1. XGBoost Feature Importance (Gain)

The table below details the statistical feature importances calculated by the XGBoost challenger model based on fractional information gain. A higher gain implies the feature is more critical in generating decision tree splits.

Feature Name	Relative Importance (Gain)	Risk Rank
inq_last_6mths	0.2513	#1
purpose	0.1737	#2
revol_util	0.1656	#3
pub_rec	0.1652	#4
annual_inc	0.1255	#5
pub_rec_bankruptcies	0.1187	#6

2. Global Risk Drivers: SHAP Summary Explanation

The chart below illustrates the SHAP (SHapley Additive exPlanations) values for the model variables. Each point represents an individual applicant. Red indicates higher feature values, while blue indicates lower values. Positive SHAP values indicate an increased probability of default.



Interpretation of Key Risk Drivers:

- **Revolving Utilization (`revol_util`):** High utilization is the strongest risk driver of default, pushing predicted PD upwards.
- **Inquiries (`inq_last_6mths`):** Recent credit inquiries represent credit-seeking behavior and increase default risk.
- **Annual Income (`annual_inc`):** Lower annual income shows negative SHAP values (reduced risk as income increases).